

Confidence Calibration Test Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and the Knower • 20-25 min

Category	Knowledge and the Knower	Time	20-25 min
Difficulty	Easy	ID	confidence-calibration-test

Big idea: Certainty and accuracy are not the same.

Overview

Students compare how confident they feel with how accurate they actually are.

Student Prompt

Answer each question, then mark your confidence: 50%, 60%, 70%, 80%, 90%, or 100%.

Concrete Stimulus Packet

Student Prompt	Answer each question privately. After marking, calculate accuracy within each confidence band.	Do not allow lookup; the point is calibration, not trivia mastery.
Question 1	Which planet is closest to the Sun?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 2	What is the capital city of Canada?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 3	Which is heavier: a kilogram of feathers or a kilogram of steel?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 4	In which year did the Berlin Wall fall?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 5	What is the chemical symbol for potassium?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 6	Which country has the longest coastline in the world?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 7	How many bones are in the adult human body?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 8	What is the only mammal capable of true sustained flight?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 9	Which has more letters: the Greek alphabet or the modern English alphabet?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 10	Without looking it up, what is the 19th digit after the decimal point in pi?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	Which planet is closest to the Sun?	Mercury
2	What is the capital city of Canada?	Ottawa
3	Which is heavier: a kilogram of feathers or a kilogram of steel?	Neither — they weigh the same
4	In which year did the Berlin Wall fall?	1989
5	What is the chemical symbol for potassium?	K
6	Which country has the longest coastline in the world?	Canada
7	How many bones are in the adult human body?	206
8	What is the only mammal capable of true sustained flight?	Bat
9	Which has more letters: the Greek alphabet or the modern English alphabet?	English alphabet — 26 vs Greek 24
10	Without looking it up, what is the 19th digit after the decimal point in pi?	4

Teacher Key / Reveal

Where were we most confident but least accurate, and what does that reveal about certainty as a way of knowing?

- Q1: Mercury
- Q2: Ottawa
- Q3: Neither — they weigh the same
- Q4: 1989
- Q5: K
- Q6: Canada
- Q7: 206
- Q8: Bat
- Q9: English alphabet — 26 vs Greek 24
- Q10: 4

Setup Checklist

- Prepare a starter stimulus using: Ten general knowledge questions with confidence bands from 50% to 100%.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Ten general knowledge questions with confidence bands from 50% to 100%.
- A mini quiz mixing easy, medium, and impossible questions so overconfidence becomes visible.
- A class calibration chart comparing confidence level with actual accuracy.

Suggested Timing

Stage	Teacher move	Watch for
1	Give students general knowledge questions. For each answer they also write a confidence level: 50%, 60%, 70%, 80%, 90% or 100%.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Mark the answers.	Assumptions, confidence shifts, evidence claims, and language choices.
3	For each confidence band, calculate actual percentage correct.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Ask students whether they were overconfident, underconfident or well calibrated.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat perception or memory as a recording device.
- Students may assume confidence is evidence rather than a feeling about evidence.

Assessment Look-Fors

- Separates perception, memory, belief, confidence, and justification.
- Uses the activity result to make a cautious claim about knowers.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Change Blindness Starter
- Personal Knowledge Map
- Blind Spot Autobiography